

SORTAVALA, Russian Federation

WMO: 228020

Lat: 61.7140N

Lon: 30.7133E

Elev: 19

StdP: 101.10

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-25.9	-22.7	-28.6	0.3	-25.8	-25.2	0.4	-22.5	8.8	-2.9	7.9	-2.9	0.8	320	0.589

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.0	26.2	19.0	24.5	18.1	22.8	17.0	20.4	24.4	19.1	22.8	18.1	21.7	2.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	13.7	22.5	17.7	12.7	21.2	16.5	11.8	20.1	58.5	24.4	54.5	22.8	51.1	21.7	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.3	6.2	5.4	-28.7	28.4	4.0	2.1	-31.5	29.9	-33.9	31.1	-36.1	32.3	-39.0	33.9		
(5)			-28.8	21.6	4.0	1.6	-31.7	22.7	-34.0	23.7	-36.2	24.6	-39.1	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.2	-7.6	-7.9	-3.5	2.7	9.0	13.8	17.4	15.4	10.4	4.7	-0.4	-4.5		
	DBStd	9.88	7.38	7.14	4.84	3.72	4.09	3.43	3.11	2.92	3.29	3.79	4.57	6.62		
	HDD10.0	2724	544	501	419	221	69	6	0	1	34	168	313	449		
	HDD18.3	5201	803	734	677	469	291	141	55	98	237	424	563	707		
	CDD10.0	607	0	0	0	2	37	121	228	169	47	3	0	0		
	CDD18.3	41	0	0	0	0	1	6	25	9	0	0	0	0		
	CDH23.3	277	0	0	0	0	8	55	162	52	0	0	0	0		
	CDH26.7	33	0	0	0	0	0	8	23	2	0	0	0	0		
(14) Wind	WSAvg	2.3	2.4	2.3	2.4	2.2	2.4	2.2	2.1	2.0	2.1	2.3	2.6	2.6		
(15) Precipitation	PrecAvg	633	44	32	31	34	39	55	61	79	67	67	67	60		
	PrecMax	892	114	101	67	65	120	138	177	156	135	144	124	143		
	PrecMin	410	7	5	6	6	3	10	11	22	15	10	22	12		
	PrecStd	120	24	19	14	16	24	27	34	32	33	37	26	27		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.8	4.2	9.0	17.1	24.4	27.7	29.0	26.8	21.3	13.7	9.0	7.0		
		MCWB	2.5	2.3	4.4	11.0	15.2	20.0	21.0	19.9	16.2	11.7	8.0	5.8		
	2%	DB	2.7	2.7	6.0	14.0	21.4	24.8	26.9	24.9	18.6	12.2	7.7	5.1		
		MCWB	1.9	1.6	2.7	8.8	13.8	17.5	19.5	18.4	14.8	10.8	7.0	4.4		
	5%	DB	1.8	1.7	3.8	11.2	18.8	22.6	25.1	22.9	16.9	11.2	6.7	3.7		
		MCWB	1.2	0.9	1.4	6.8	12.1	16.2	18.9	17.2	13.8	9.9	6.0	3.0		
	10%	DB	1.0	0.9	2.5	8.9	16.5	20.7	23.5	21.2	15.5	10.1	5.6	2.7		
		MCWB	0.5	0.3	0.8	5.1	10.9	15.1	17.8	16.7	13.1	9.0	5.0	1.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.0	2.7	4.9	11.5	17.2	21.0	22.6	21.3	16.8	12.4	8.3	6.0		
		MCDB	3.4	3.4	8.1	15.6	21.7	26.5	27.4	24.8	19.2	13.2	8.8	6.7		
	2%	WB	2.0	1.7	3.1	9.2	14.7	18.9	20.9	19.4	15.4	11.3	7.0	4.5		
		MCDB	2.5	2.4	5.3	13.1	19.2	23.5	25.0	23.1	17.6	12.0	7.6	5.0		
	5%	WB	1.3	1.1	1.9	7.3	13.0	17.2	19.8	18.2	14.4	10.2	6.1	3.1		
		MCDB	1.7	1.6	3.2	10.9	18.0	21.1	23.5	21.9	16.2	10.9	6.6	3.6		
	10%	WB	0.6	0.4	1.1	5.4	11.5	15.7	18.7	17.1	13.6	9.1	5.0	2.1		
		MCDB	1.0	0.9	2.1	8.3	15.6	19.8	22.0	20.2	15.2	9.9	5.5	2.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	6.3	7.4	8.2	9.2	8.6	8.0	8.0	6.7	4.7	3.8	4.6		
		MCDBR	3.5	4.7	8.4	12.5	13.2	11.8	10.7	11.0	8.8	5.6	4.0	3.5		
	5% WB	MCWBR	3.4	4.4	6.0	7.6	7.0	5.7	4.9	5.3	5.2	4.3	3.9	3.4		
		MCDWR	3.6	4.3	6.6	11.7	11.4	9.9	9.2	9.2	7.2	4.9	3.8	3.2		
(40) Clear-Sky Solar Irradiance	taub	taub	0.250	0.297	0.323	0.361	0.359	0.352	0.381	0.366	0.341	0.322	0.270	0.220		
		taud	2.036	2.093	2.176	2.316	2.438	2.486	2.422	2.448	2.503	2.396	2.126	1.977		
	Edn at Noon	Edn at Noon	428	645	778	826	863	875	835	818	774	640	404	281		
		Edh at Noon	39	73	96	103	99	97	101	91	72	56	35	23		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.21	0.77	2.25	3.70	5.05	5.43	5.25	4.02	2.29	0.89	0.22	0.11		
	RadStd	RadStd	0.03	0.19	0.31	0.39	0.51	0.33	0.46	0.46	0.29	0.13	0.04	0.01		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.57	N/A	N/A	N/A	N/A	N/A	-214	N/A	N/A	N/A
(47)	Regional (20 neighbors)	+0.54	N/A	N/A	N/A	N/A	N/A	-200	N/A	N/A	N/A

Nomenclature: See separate page